

case, the simulation includes 100 switch controllers with dead zone, 100 transport delays, and 200 second-order systems, each with a natural frequency of 195 Hz.

Note that the modified Euler integration method and the multiple framing for the discontinuous switch function represent techniques that can improve the dynamic accuracy of any digital simulation.

2. Description of the Control System

Figure 1 shows a block diagram of the overall satellite control system.

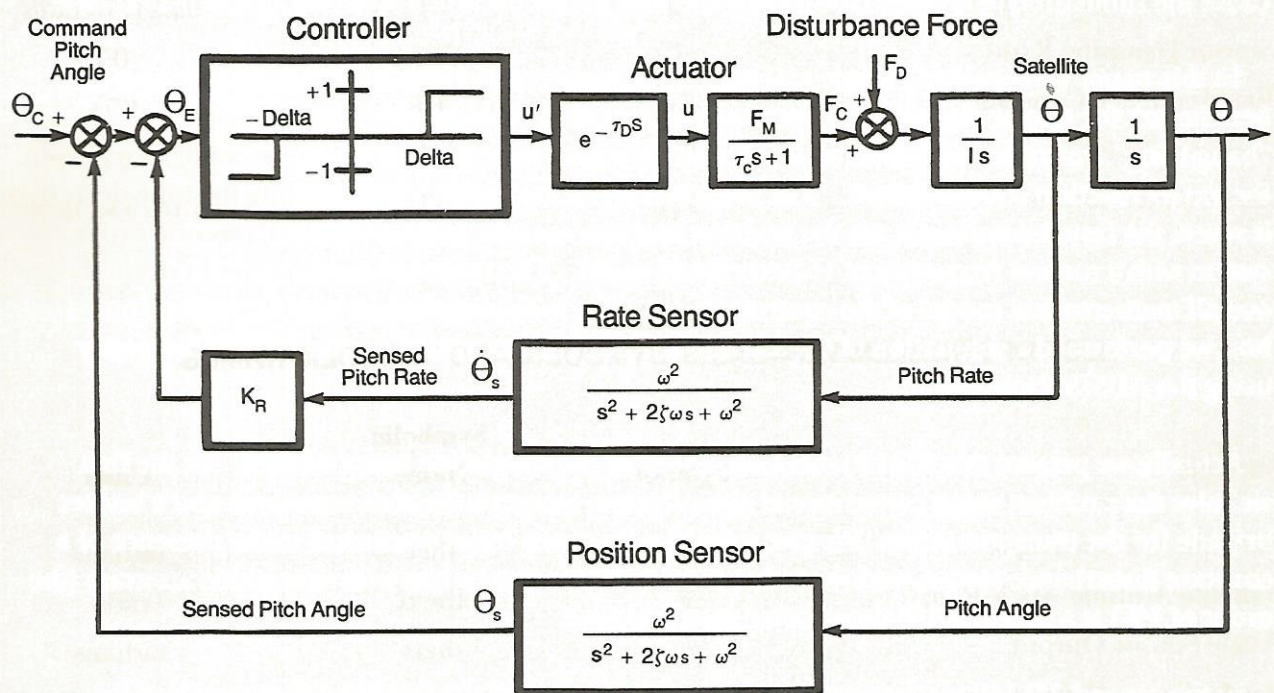


Figure 1
A Satellite Attitude Control System

Table 1 lists the specific parameter values used in the simulation. The initial conditions for all state variables, including the delayed controller output, u , are equal to 0. Table 2 lists the state variables and other problem variables along with the symbolic names used in the ADSIM program.

TABLE 1
PARAMETER NAMES AND VALUES

Parameter	Symbol	Symbolic Name	Value
Command Pitch Angle	θ_c	THETAC	0.09 radian
Dead Zone HalfWidth	delta	DELTA	0.008 radian
Time Delay	τ_D	TD	0.025 second
Actuator Time Constant	τ_c	TC	0.050 second
Spacecraft Inertia	I	I	100 kg m ²
Sensor Natural Frequency	ω	OMEGA	94.24778 rad/s
Sensor Damping Ratio	ζ	ZETA	0.7
Rate Feedback Constant	K _R	KR	0.5 s
Actuator Torque	F _M	FM	25 kg m ² /s ²
Disturbance Torque	F _D	FD	12.1875 kg m ² /s ²

TABLE 2
LIST OF PROBLEM VARIABLES, SYMBOLS, AND SYMBOLIC NAMES

Variable	Symbol	Symbolic Name	Units
Satellite Attitude Angle	θ	thet	radians
Satellite Attitude Angle Rate	$\dot{\theta}$	thetd	rad/s
Angle Sensor Output	θ_S	thets	radians
Angle Sensor Output Derivative	$\dot{\theta}'_S$	thetp	rad/s
Rate Sensor Output	$\dot{\theta}_S$	thetd	rad/s
Rate Sensor Output Derivative	$\dot{\theta}'_S$	thetdp	rad/s ²
Error	θ_E	thete	radians
Controller Switch Output	u'	up	
Delayed Switch Output	u	u	
Actuator Output	F _C	f	kg m ² /s ²